

Fake Job Posting Detection Using Explainable NLP Models

User Manual

Prepared by: Heng Xin Phei (23005228)



This system is a web-based application built using Streamlit that helps users identify whether a job posting is Real or Fake. It uses a trained TF-IDF + XGBoost model combined with structured job features and provides Explainable AI (XAI) using LIME so users can understand why a prediction is made.

1

INTRODUCTION



2

SYSTEM REQUIREMENTS

Users only need a device with an internet connection and a web browser such as Google Chrome, Microsoft Edge, Firefox, or Safari.



3

APPLICATION STRUCTURE

The application contains two main tabs:

1. Single Job Check
2. Bulk CSV Detection

4

HOW TO LAUNCH THE APPLICATION

Open the [application link](https://fake-job-detector-dwc2pa8ngwsczqb35ccvqx.streamlit.app/) using any web browser, and the application will be ready for use.



<https://fake-job-detector-dwc2pa8ngwsczqb35ccvqx.streamlit.app/>

A

SINGLE JOB CHECK TAB

This feature allows users to test one job posting at a time.

The screenshot shows the 'Fake Job Posting Detector' web application. It has a header with a detective icon and the title 'Fake Job Posting Detector'. Below the title is a subtitle 'AI-powered job scam detection' and a note 'Assess whether a posting is Real or Potentially Fake.' A red italicized note says 'The more information you provide, the more precise our analysis will be.' There are two tabs: 'Single Job Check' (selected) and 'Bulk CSV Detection'. The form is divided into two columns. The left column contains input fields for: Title (filled with 'Administrative Assistant'), Location (filled with 'US, CA, Santa Ana'), Department (filled with 'Administration'), Salary Range (filled with '\$1000-\$3000'), Employment Type (filled with 'Full-time'), Required Experience (filled with '2 years'), Required Education (filled with 'bachelor degree'), Industry (filled with 'Accounting'), and Function (filled with 'Administrative'). The right column contains: Company Profile (filled with 'Information not provided by employer'), Description (filled with 'In addition to clerical and administrative duties, individual will be asked to support the office staff in other areas.- Individual must have strong math skills.- Drug Screen'), Requirements (empty), and Benefits (filled with 'No requirements'). At the bottom right are three toggle switches: 'Remote / Telecommuting' (checked), 'Company Logo Provided' (unchecked), and 'Screening Questions Included' (unchecked). At the bottom left is a checkbox 'Show Explanation' and a 'Predict' button. Red dashed boxes highlight the Title, Description, and Show Explanation fields. Red dashed arrows point from these boxes to numbered callouts on the right. A blue bracket groups the three toggle switches, with a blue arrow pointing to callout 3. A green dashed arrow points from callout 4 to the Predict button.

1 Enter job title and description

2 Fill optional text and structured fields

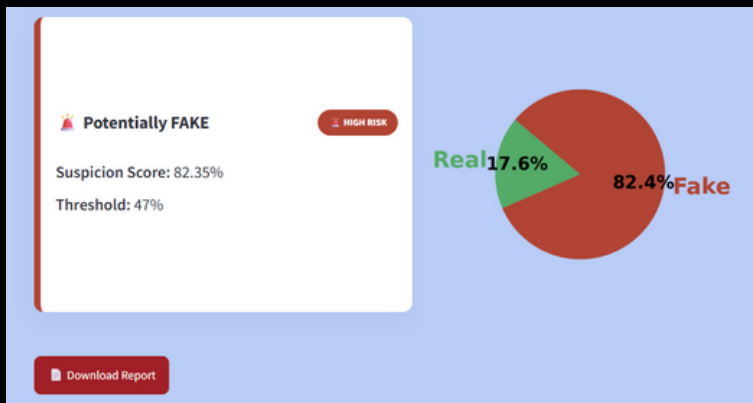
3 Toggle the button to show

- remote / telecommuting
- company logo
- screening questions

4 Click **Predict**

User can tick the **Show Explanation** box for explainability and transparency (LIME)

Output



5 Click Download Report

If user tick **SHOW EXPLANATION**

The system will display:

- Prediction result:
 - Real Job Posting
 - Fake Job Posting
- Fraud probability score (threshold = 0.47)
- Decision based on optimized threshold

Fake Job Posting Detection Report	
=====	
FINAL DECISION: POTENTIALLY FAKE	
Suspicion Score: 82.35%	
Risk Alert Level: 47%	
SUBMITTED JOB DETAILS:	
TITLE: Administrative Assistant	
LOCATION: US, CA, Santa Ana	
DEPARTMENT: Administration	
SALARY_RANGE: \$1000-\$3000	
EMPLOYMENT_TYPE: Full-time	
REQUIRED_EXPERIENCE: 2 years	
REQUIRED_EDUCATION: bachelor degree	
INDUSTRY: Accounting	
FUNCTION: Administrative	
COMPANY_PROFILE: Information not provided by employer	
DESCRIPTION: In addition to clerical and administrative duties, individual will be asked to support...	
REQUIREMENTS:	
TELECOMMUTING: 1	
HAS_COMPANY_LOGO: 0	
HAS_QUESTIONS: 0	
FRAUDULENT: 0	
TOP INFLUENTIAL KEYWORDS:	
• areas: supports Real Weight: 0.1499	
• information: supports Fake Weight: 0.1429	
• Assistant: supports Fake Weight: 0.0913	
• must: supports Real Weight: 0.0901	
• administrative: supports Fake Weight: 0.0550	
• Administrative: supports Fake Weight: 0.0541	
• strong: supports Real Weight: 0.0433	
• office: supports Fake Weight: 0.0266	
• addition: supports Fake Weight: 0.0211	
• clerical: supports Fake Weight: 0.0182	

in pdf format

1

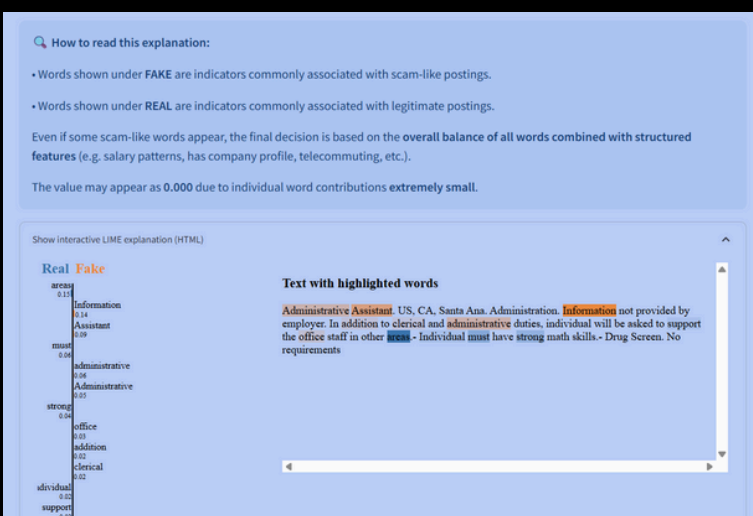
List of words supporting REAL and FAKE

Why did the model make this decision?	
Note: LIME weights are not probabilities. They show which words push the model locally. Even if some words look scam-like, the final decision is based on the overall balance + structured features.	
Words supporting REAL	Words supporting FAKE
• areas (0.150) • must (0.060) • strong (0.043) • Individual (0.018) • support (0.017)	• Information (0.143) • Assistant (0.091) • administrative (0.055) • Administrative (0.054) • office (0.027) • addition (0.021) • clerical (0.018)

2

LIME explanation showing:

- Words that push prediction towards Fake (orange)
- Words that push prediction towards Real (blue)



B

BULK CSV DETECTION TAB

This feature is used to analyze multiple job postings at once.

Fake Job Posting Detector

AI-powered job scam detection

Assess whether a posting is Real or Potentially Fake.

The more information you provide, the more precise our analysis will be.

Single Job Check **Bulk CSV Detection**

Bulk CSV Detection

Upload a CSV containing job details to process multiple entries at once.

Download CSV Template

Important CSV Requirements:

- **Location Format:** Country, State, City (e.g., US, NY, New York)
- **Binary Columns:** use 1 for Yes and 0 for No
- **No Column Name Changes:** follow the template

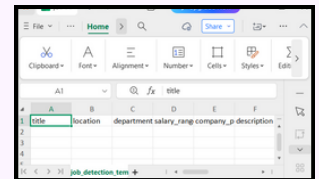
Upload CSV

Drag and drop file here
Limit 200MB per file • CSV

Browse files

bulk csv detection.csv 1.5KB

Run Bulk Analysis



1

Click **Download CSV Template**

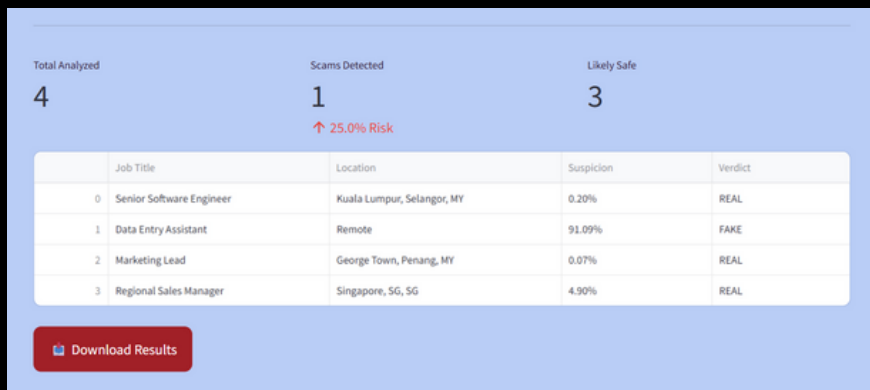
2

Upload CSV files, ensure it meets all requirements

3

Click **Run Bulk Analysis**

Output



The system will:

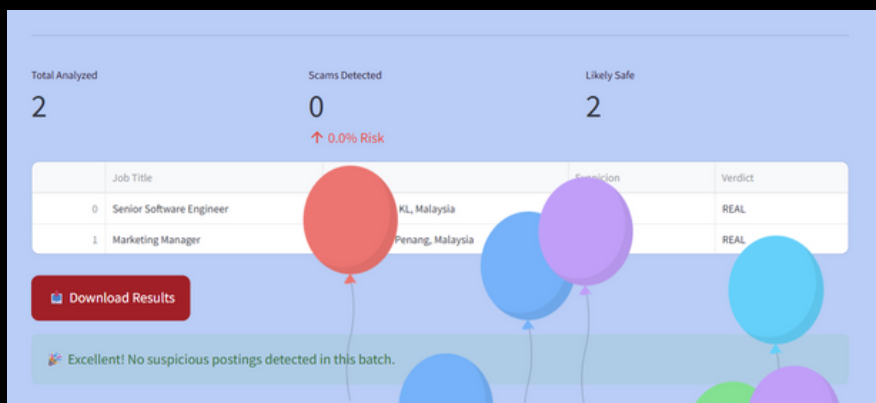
- Analyze all job postings
- Display total number of analyzed, scams detected and likely safe postings
- Show the suspicion percentage and verdict (REAL or FAKE)

4 Click Download Results

#	A	B	C	D
1	Job Title	Location	Suspicion	Verdict
2	Senior Softv	Kuala Lump	0.20%	REAL
3	Data Entry /	Remote	91.09%	FAKE
4	Marketing L	George Tow	0.07%	REAL
5	Regional Sal	Singapore, S	4.90%	REAL
6				

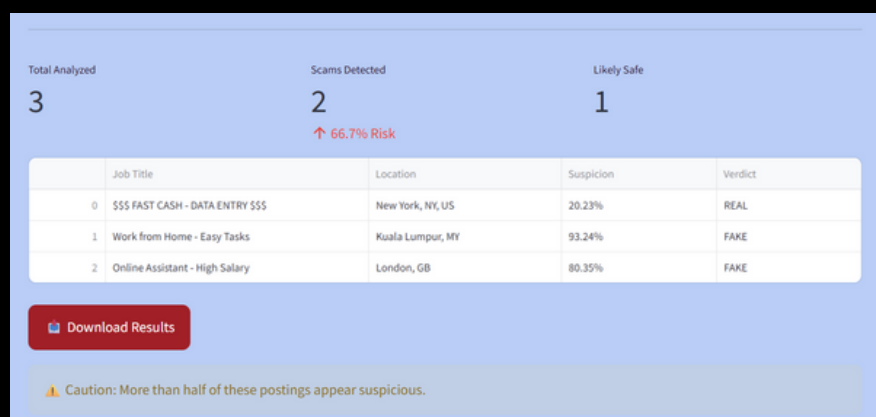
in csv format

When **all** job posting are likely safe



Balloons trigger after click **Run Bulk Analysis**

When **> half** job postings are likely fake



Warning message pop up after click **Run Bulk Analysis**